

ILPO VÄÄTÄINEN

Evaluation Integrity · AI Red-Teaming · Applied Statistics

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PROFILE

I work on evaluation integrity: finding where a benchmark number diverges from real-world performance, and building small, independently-verifiable tools to measure the gap. The same adversarial instinct drives my security work — breaking systems to see where reported behavior and real behavior part ways. Every result I report is re-derived through a second, independent code path before I believe it. I ship working software end to end and have a merged contribution to a major open-source project.

EVALUATION INTEGRITY & LLM EVALUATION

Independent research & tooling · Python · applied statistics · 2026 – present

- **calibration_audit & rank_audit** — tools that audit the numbers a leaderboard reports, not just the models it ranks. Reproject a detector's benchmark precision to its real deployment base rate, and count (null-gated) the specialization axes a single ranking hides.
- Findings, each re-verified independently: a clinical variant predictor reading **98% AUC but ~12% precision at 1% prevalence**; the AI-text detector the commercial tools descend from scoring near a **coin flip (AUC 0.55) on GPT-4**; leaderboards that are genuinely k-dimensional (MTEB k=6, ProteinGym k=3).
- **Judge / leaderboard audits**: measured LLM-as-judge bias on real data (length and self-preference effects), and audited a #1 arena ranking over tens of thousands of head-to-head battles — separating real signal from judge and sampling artifacts.
- Discipline: every figure computed from public data and re-verified through a separate implementation (e.g. AUCs cross-checked with an independent method, headline numbers carrying bootstrap intervals).

SECURITY & RED-TEAMING

Independent research · Foundry · Slither · Halmos · Immunefi / HackenProof

- DeFi vulnerability research validated end to end (unit → fuzz → invariant → Halmos symbolic → mainnet-fork) before reporting; e.g. a first-depositor share-inflation bug proven with a Foundry PoC and confirmed reachable on the deployed config.
- **Merged open-source contributions to Hummingbot** (a production trading framework), owned end to end including tests and review.

SELECTED PROJECTS

Ranger Sovereign Vault — Solana · Anchor · Python

Delta-neutral funding-arbitrage vault designed and deployed to Solana mainnet end to end: role-gated access, timelocked emergency withdrawals, Python hedge-maintenance layer. **Honorable Mention, 2026 hackathon.**

Chimera Trading Engine — Python · Hyperliquid

Live signal-driven trading automation with an LLM risk-gate, a self-built risk layer (notional caps, kill-switches) and a statistical validation pipeline (bootstrap CI, permutation tests, overfitting checks).

TECHNICAL SKILLS

Evaluation	Calibration & base-rate/precision analysis, benchmark-dimensionality auditing, judge-bias detection, red-teaming detection systems, reproducible measurement
Statistics	Bootstrap CI, permutation testing, parallel-analysis nulls, multiple-comparison correction, overfitting detection, independent re-verification
Engineering	Python (primary); building, testing & shipping tools and automation; Git/GitHub, CI, Linux; Playwright, APIs, data pipelines
Security	Foundry (unit/fuzz/invariant/mainnet-fork), Halmos, Slither; DeFi protocol analysis; Solana/Anchor, EVM/Solidity; Ed25519
Languages	Finnish (native), English (professional working proficiency)

EDUCATION

Datanomi — Vocational Qualification in Information & Communications Technology — Keuda, Kerava, Finland · 2009–2011